# Approximations

2025-10-07

## Simulated Approximations

I want to start by comparing the Normal approximation to the Binomial distribution for large $n$ and small $p$. I want to do some limit testing on exactly how big $n$ needs to be and how small $p$ needs to be relative to $n$, and compare it the rule-of-thumb that $np \geq 10$ and $n(1-p) \geq 10$.

This run will be exploratory to give me a guess for where the binomial and normal distributions start to look similar

```
ns = 10^(1:5)
ps = 10^-(seq(from=1, to=5, by=0.5))
nsamples = 1000
set.seed(54321)
for(n in ns) {
  for(p in ps) {
    mu  = n*p
    var = n*p*(1-p)

    # Skip combinations where np is too small
    if (mu < 5) next

    normal = rnorm(n=nsamples, mean=mu, sd=sqrt(var))
    binom  = rbinom(n=nsamples, size=n, prob=p)

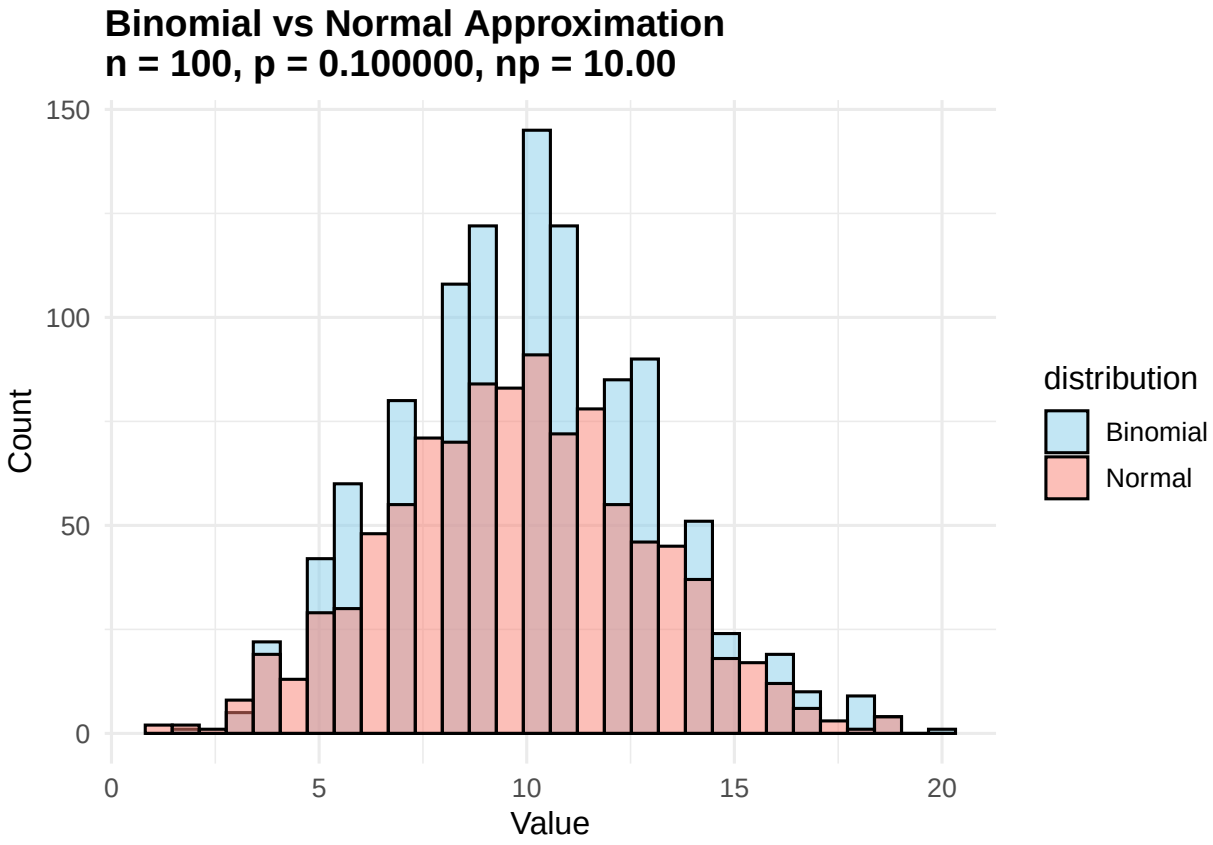
    df <- data.frame(
      value = c(binom, normal),
      distribution = rep(c("Binomial", "Normal"), each = nsamples)
    )

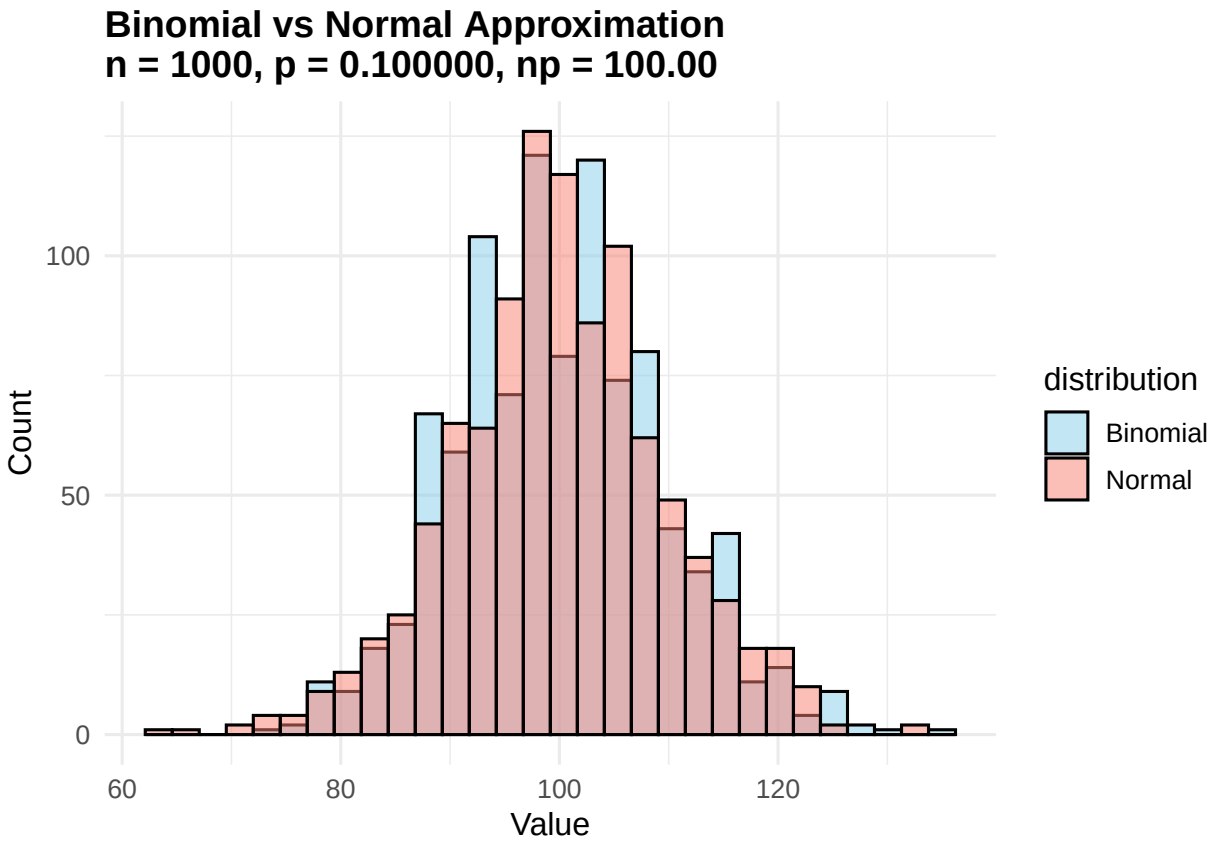
    p_title <- sprintf("n = %g, p = %.6f, np = %.2f", n, p, mu)

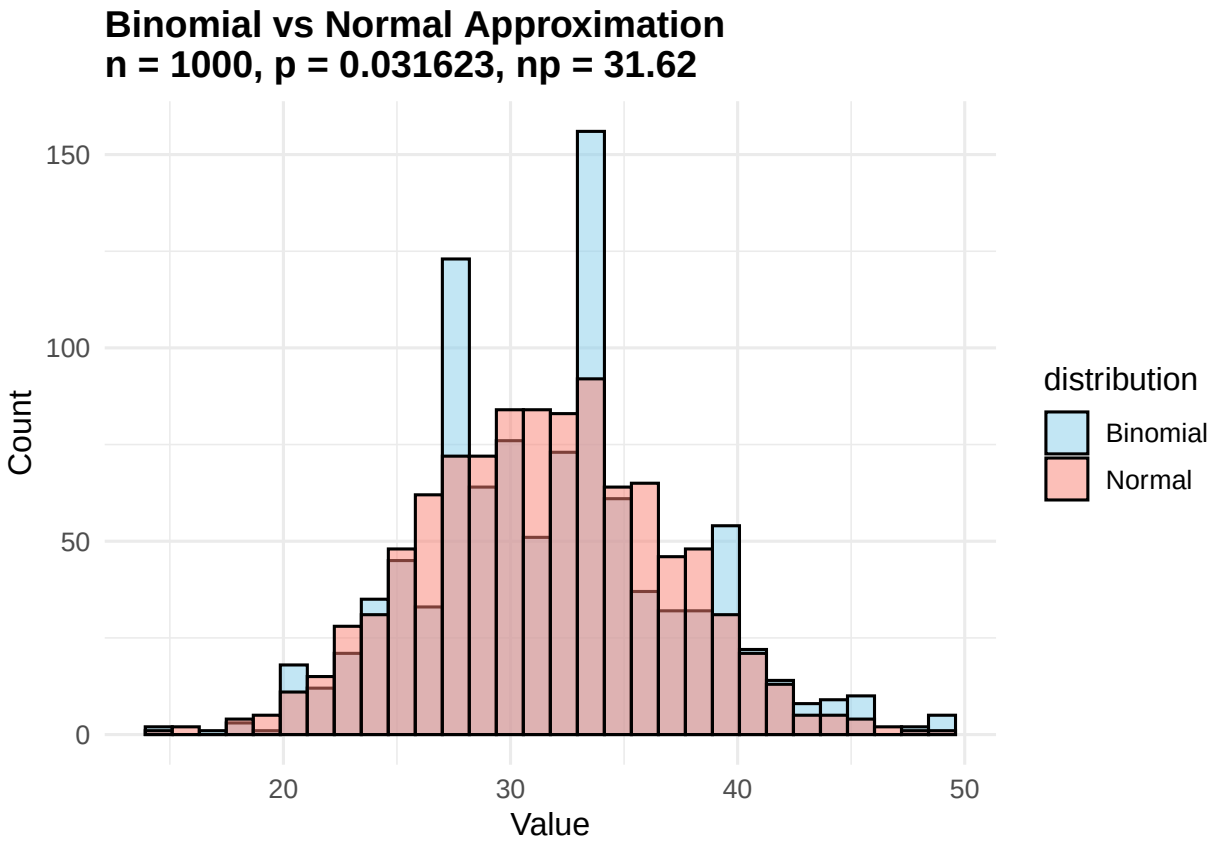
    plot <- ggplot(df, aes(x = value, fill = distribution)) +
      geom_histogram(position = "identity", bins = 30, alpha = 0.5, color = "black") +
      scale_fill_manual(values = c("Binomial" = "skyblue", "Normal" = "salmon")) +
      labs(
        title = sprintf("Binomial vs Normal Approximation\nn = %g, p = %.6f, np = %.2f", n, p, mu),
        x = "Value",
        y = "Count"
      ) +
      theme_minimal(base_size = 12) +
      theme(plot.title = element_text(size = 14, face = "bold"))

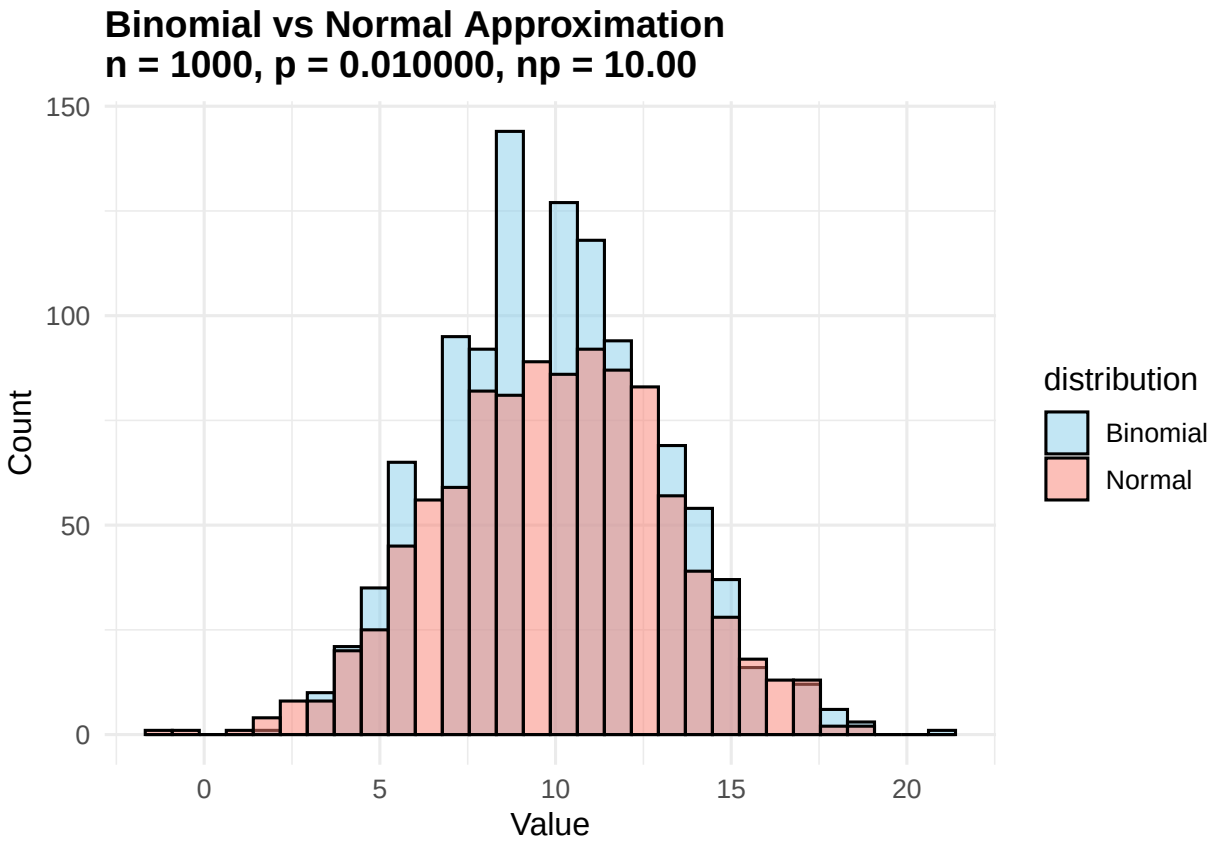
    print(plot)
    # Optional: pause for viewing
```

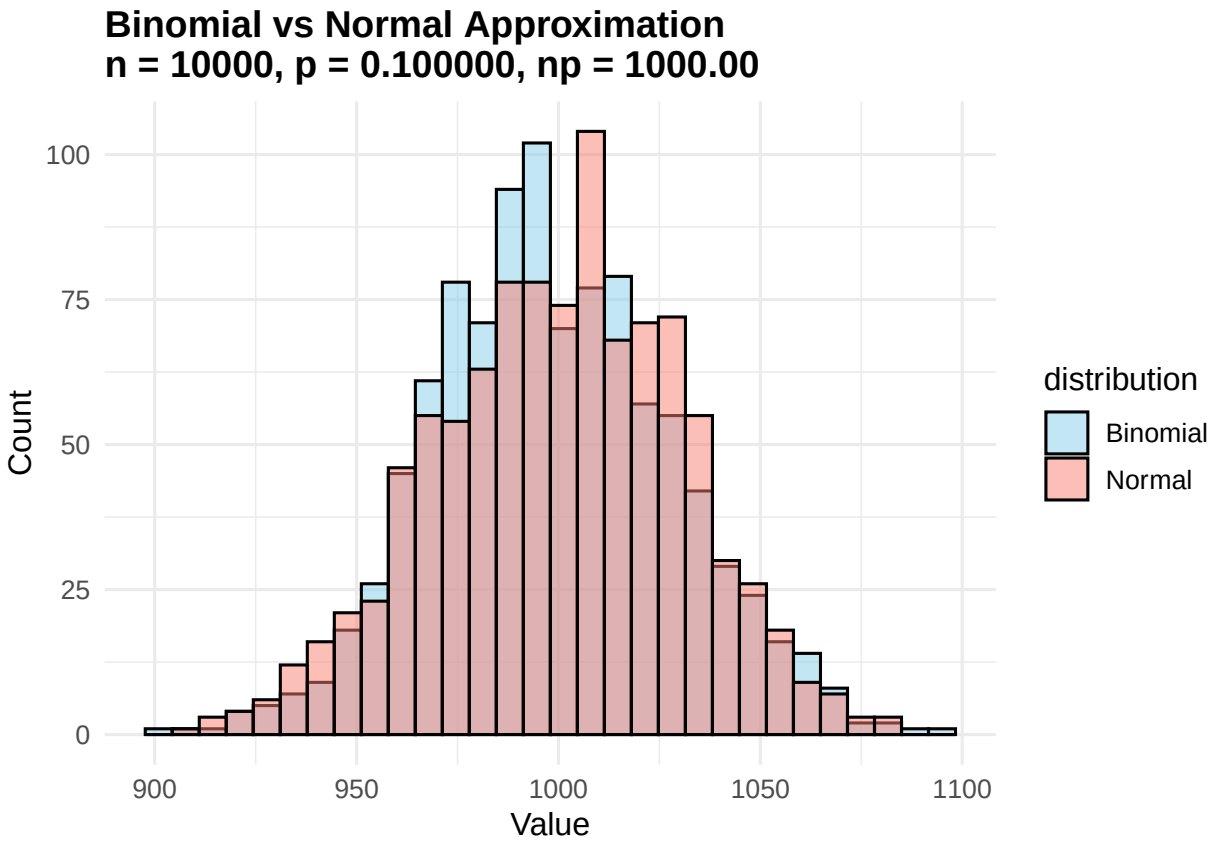
```
    # readline(prompt="Press [Enter] to continue to the next plot...")
  }
}
```

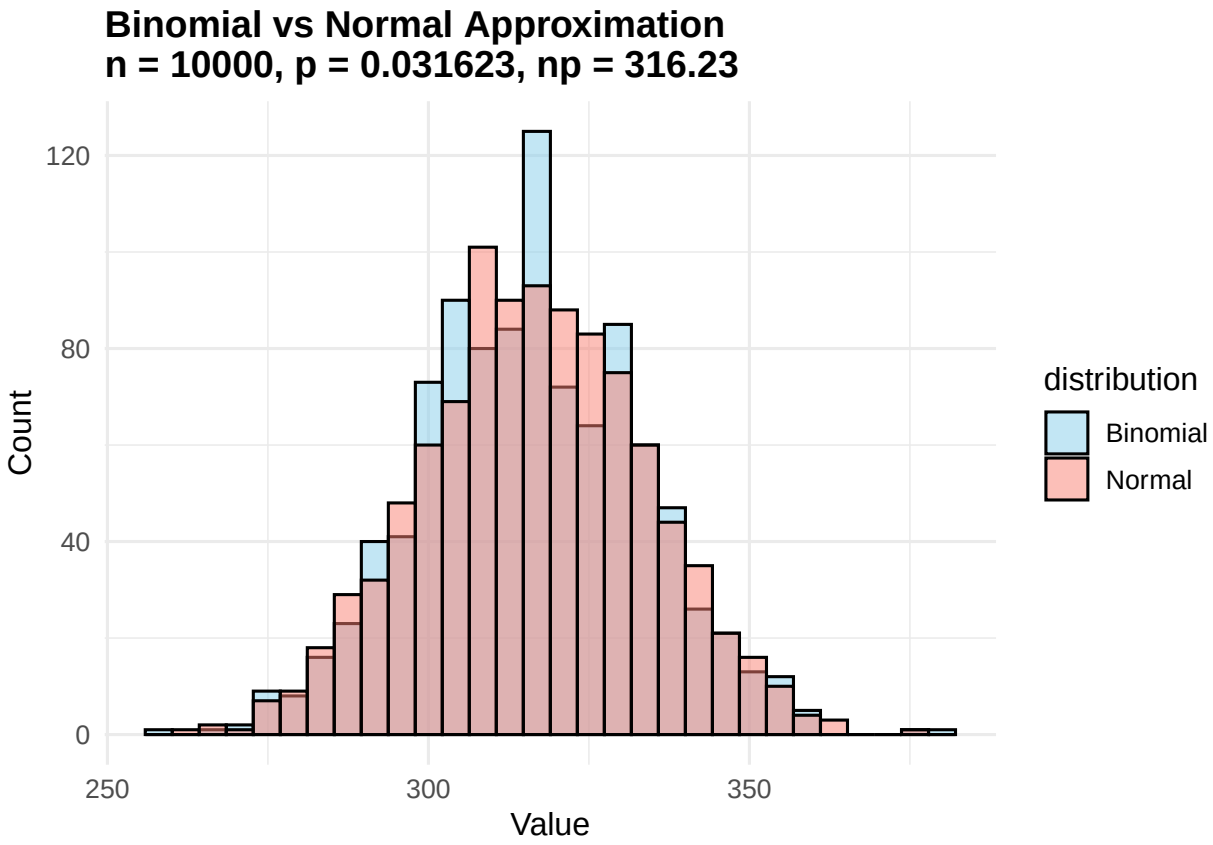


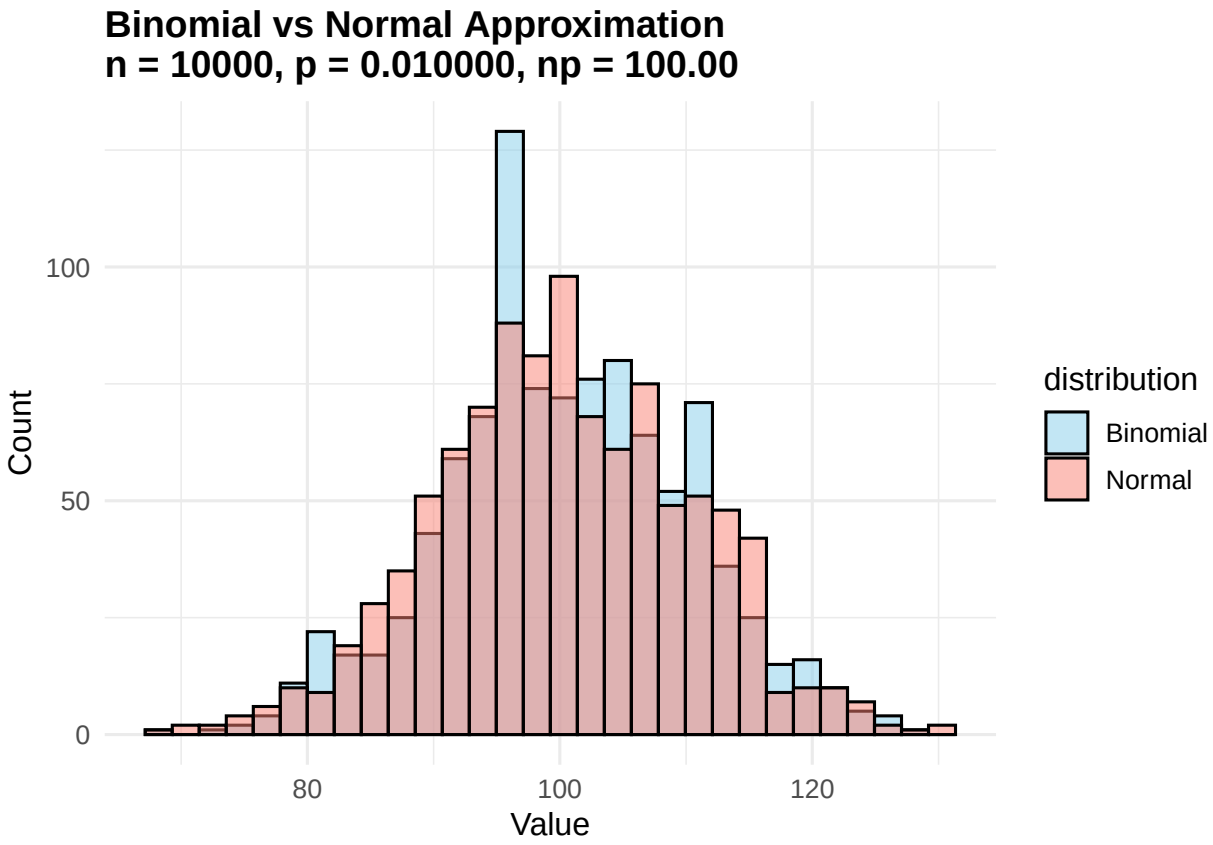

Binomial vs Normal Approximation
n = 1000, p = 0.100000, np = 100.00
Count
100
50
0
60
80
100
120
Value
distribution
Binomial
Normal

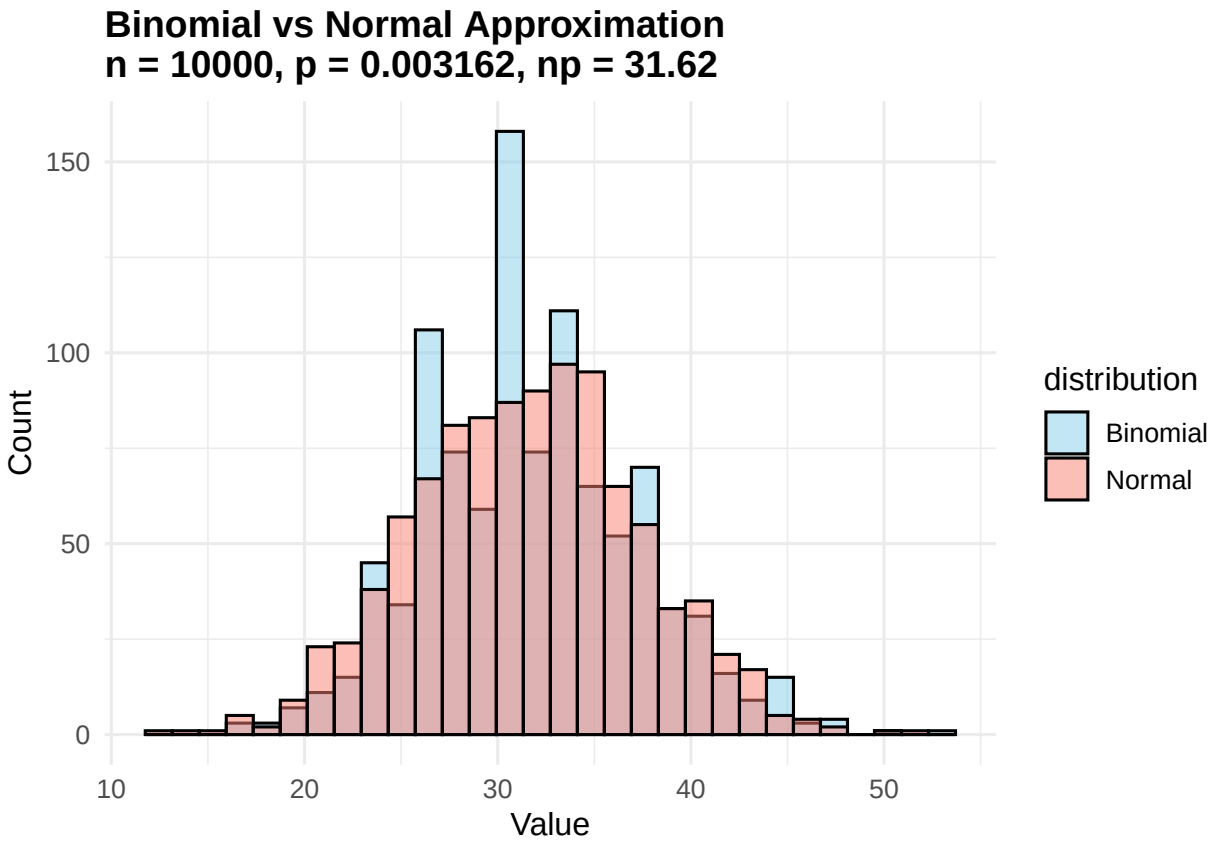

Binomial vs Normal Approximation
n = 1000, p = 0.031623, np = 31.62
Count
150
100
50
0
20
30
40
50
Value
distribution
Binomial
Normal

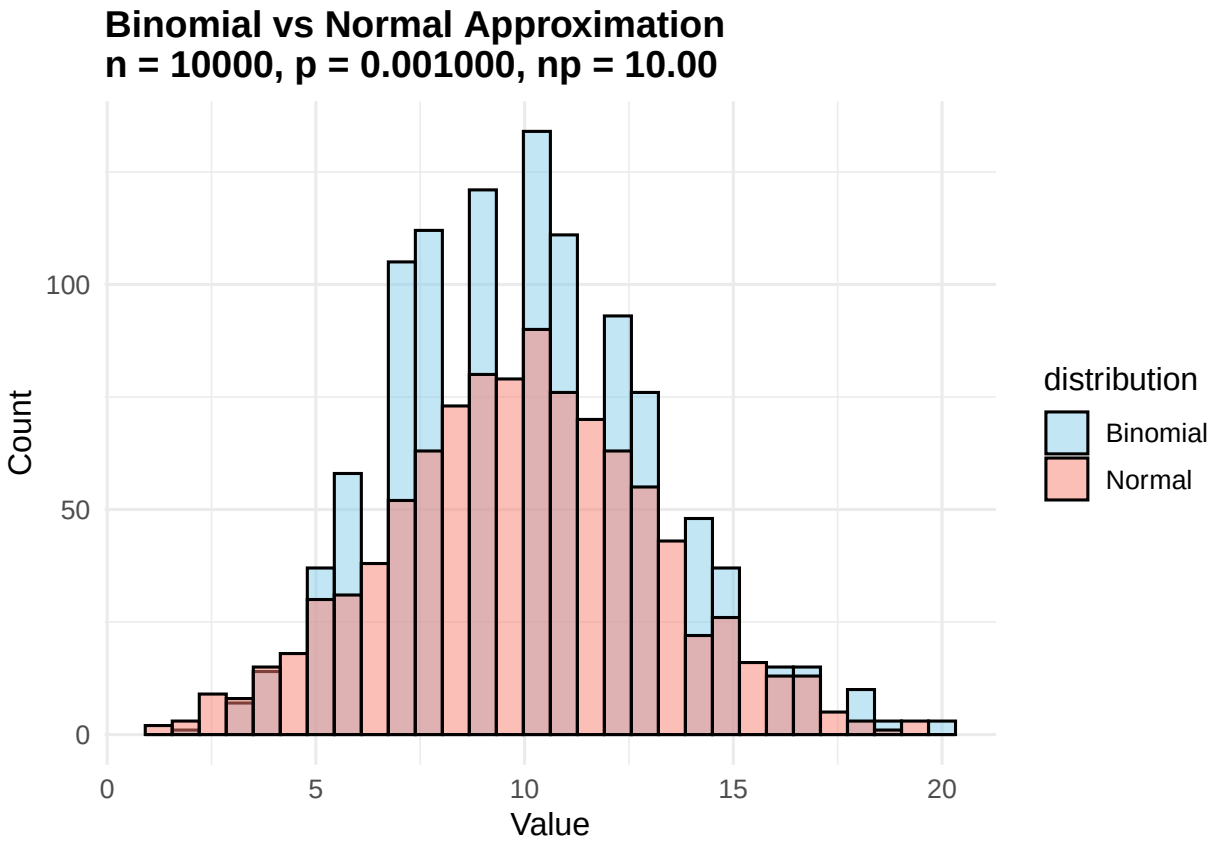

Binomial vs Normal Approximation
n = 1000, p = 0.010000, np = 10.00
Count
150
100
50
0
0
5
10
15
20
Value
distribution
Binomial
Normal

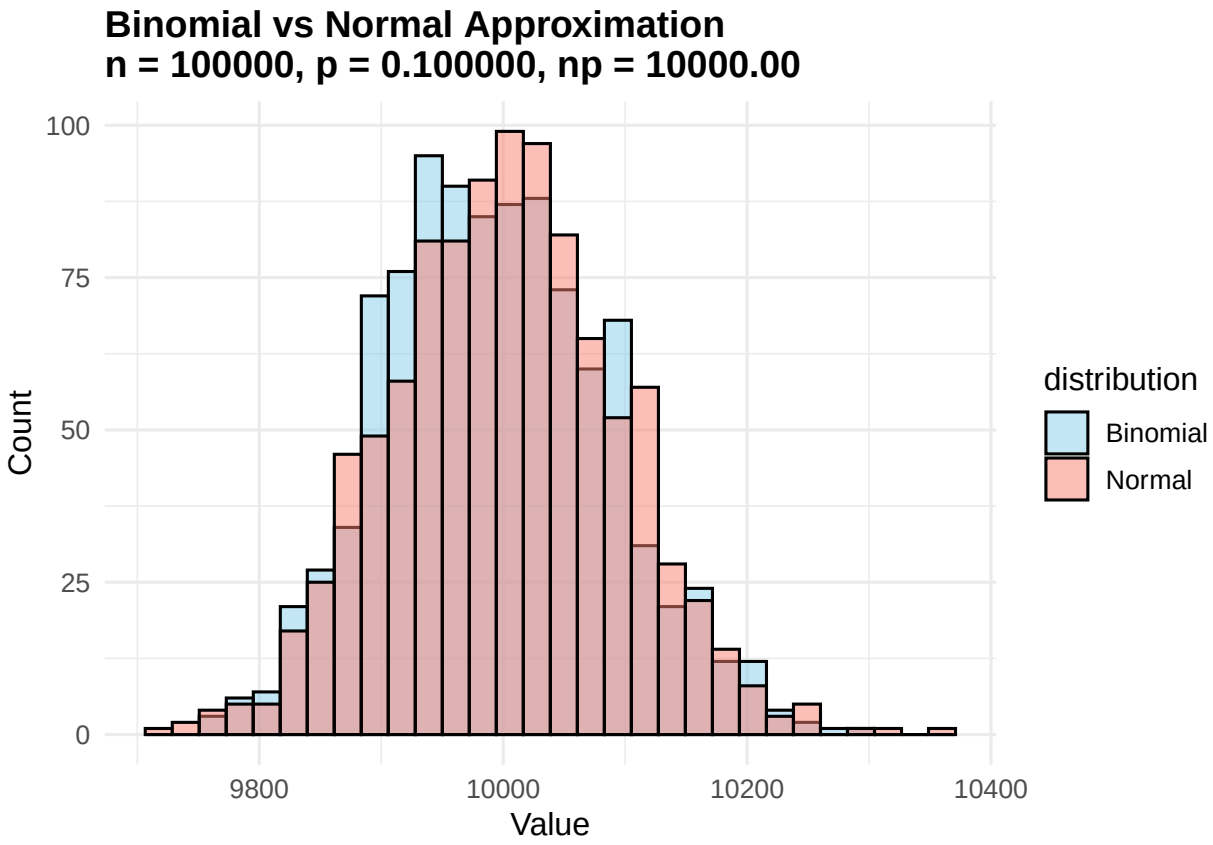

Binomial vs Normal Approximation
n = 10000, p = 0.100000, np = 1000.00
Count
Value
100
75
50
25
0
900
950
1000
1050
1100
distribution
Binomial
Normal

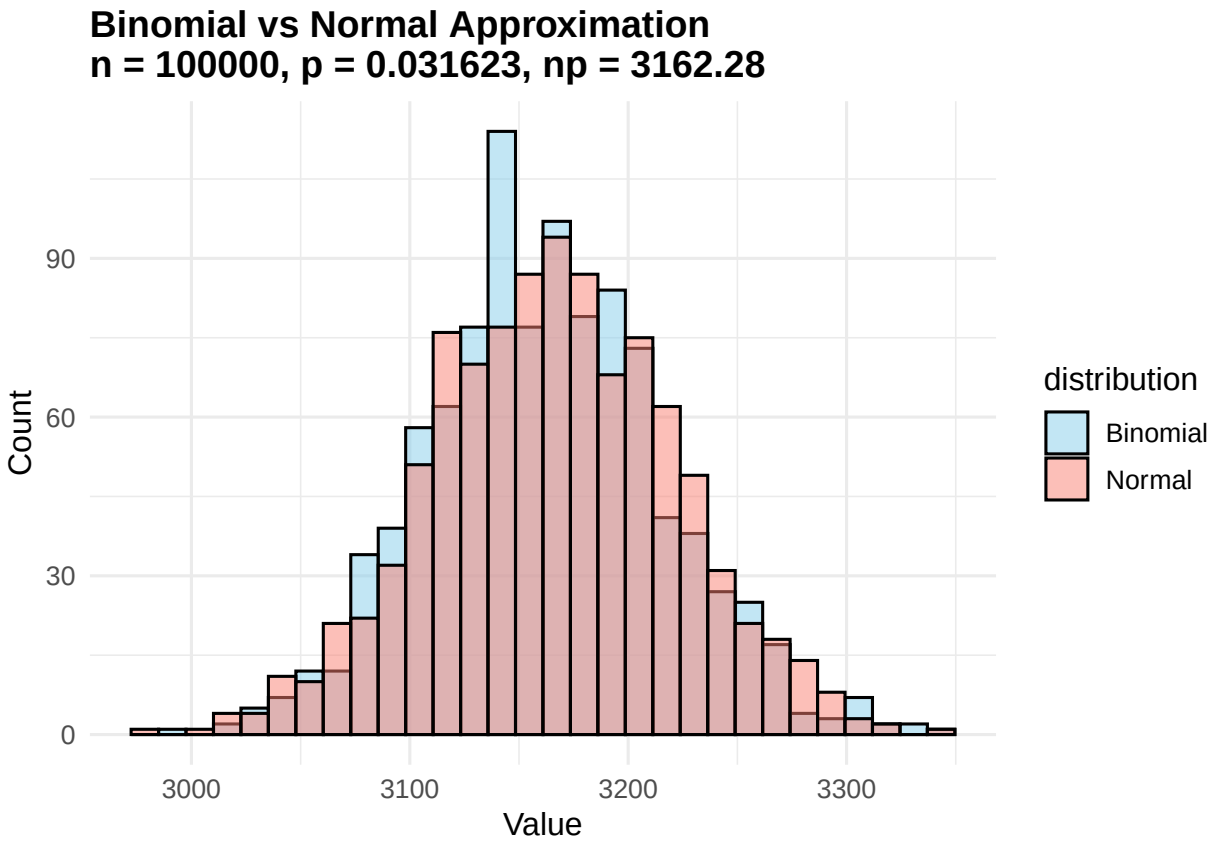

Binomial vs Normal Approximation
n = 10000, p = 0.031623, np = 316.23
Count
120
80
40
0
250
300
350
Value
distribution
Binomial
Normal

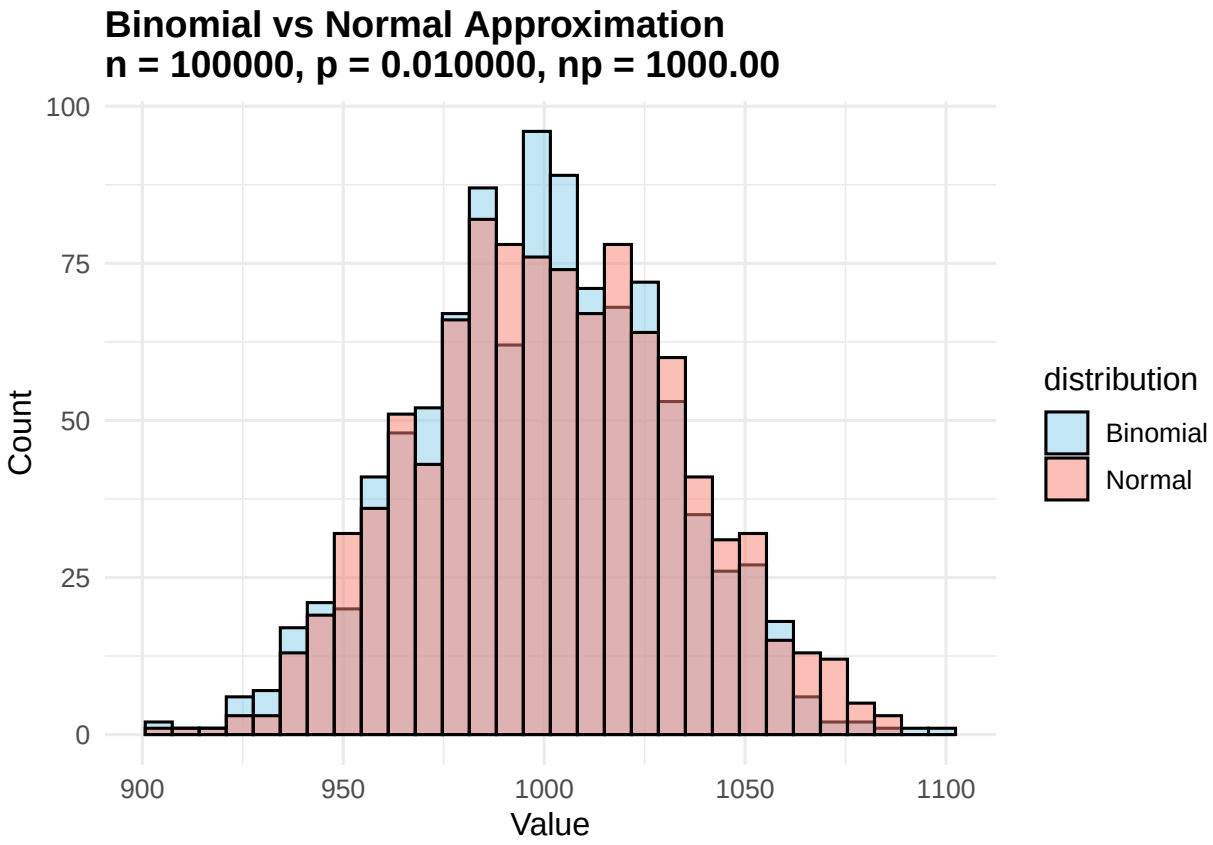

Binomial vs Normal Approximation
n = 10000, p = 0.010000, np = 100.00
Count
100
50
0
80
100
120
Value
distribution
Binomial
Normal

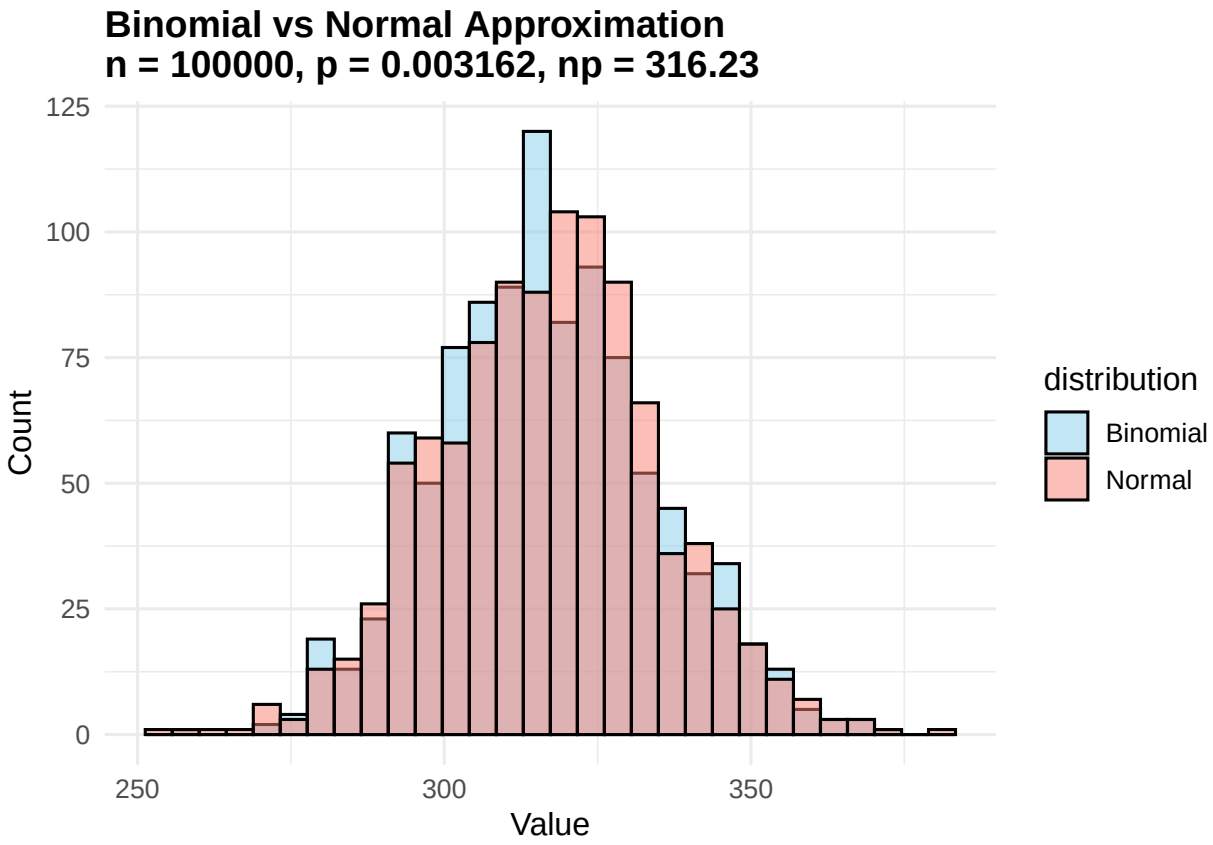

Binomial vs Normal Approximation
n = 10000, p = 0.003162, np = 31.62
Count
150
100
50
0
10
20
30
40
50
Value
distribution
Binomial
Normal

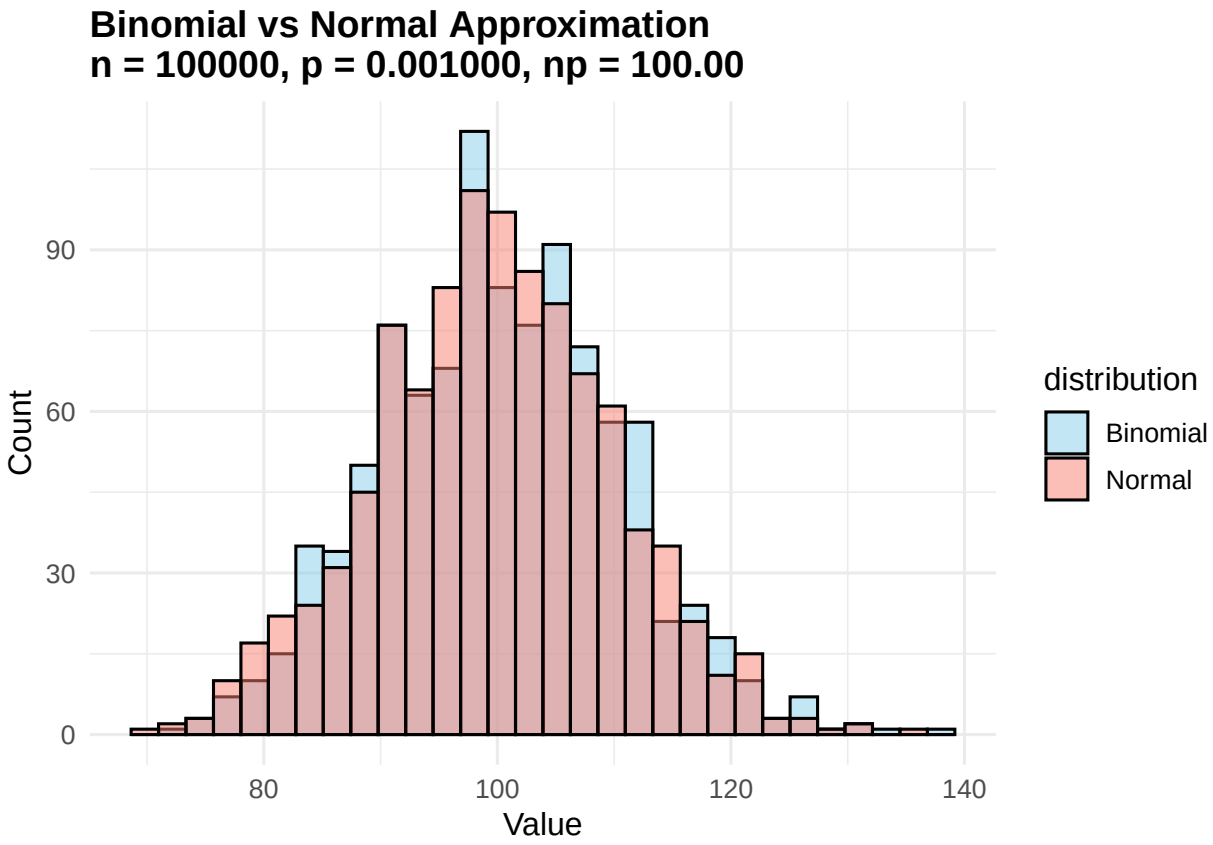

Binomial vs Normal Approximation
n = 10000, p = 0.001000, np = 10.00
Count
100
50
0
0
5
10
15
20
Value
distribution
Binomial
Normal

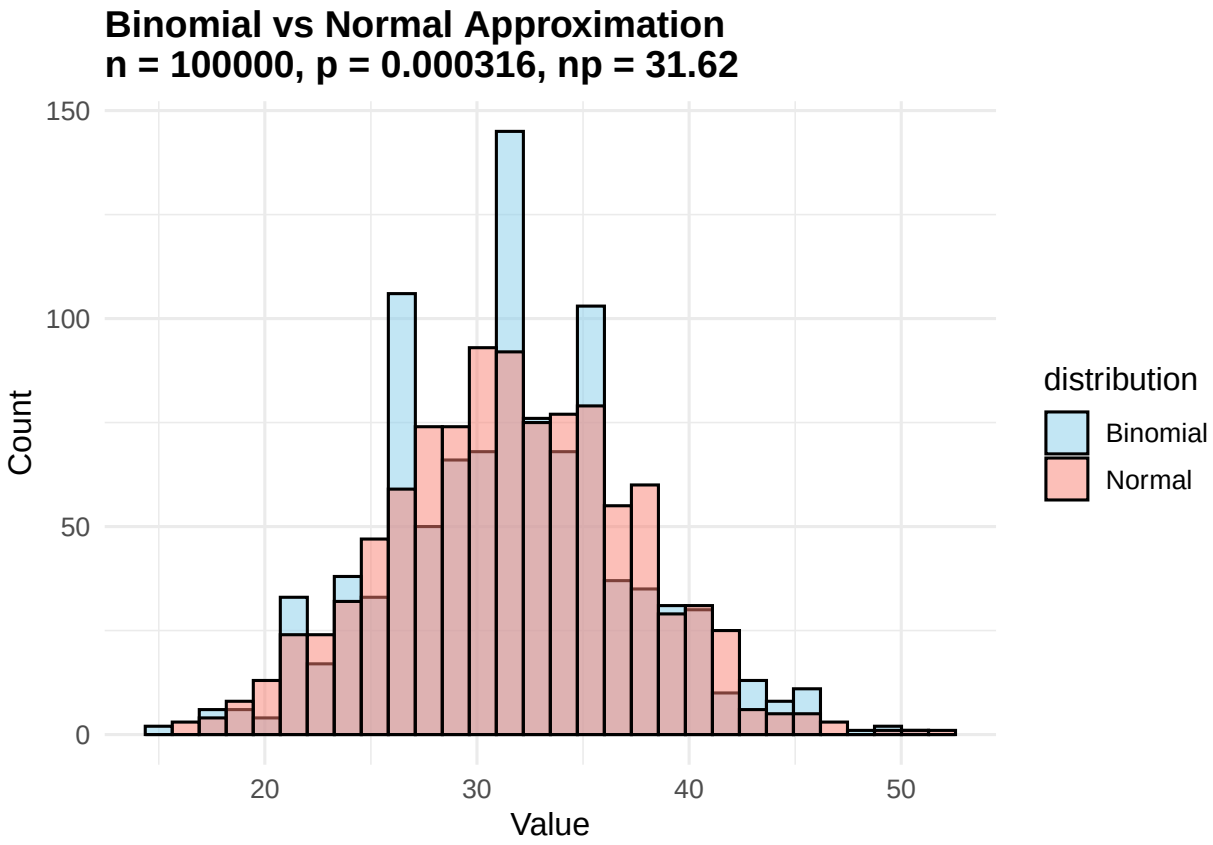

Binomial vs Normal Approximation
n = 100000, p = 0.100000, np = 10000.00
Count
100
75
50
25
0
9800
10000
10200
10400
Value
distribution
Binomial
Normal

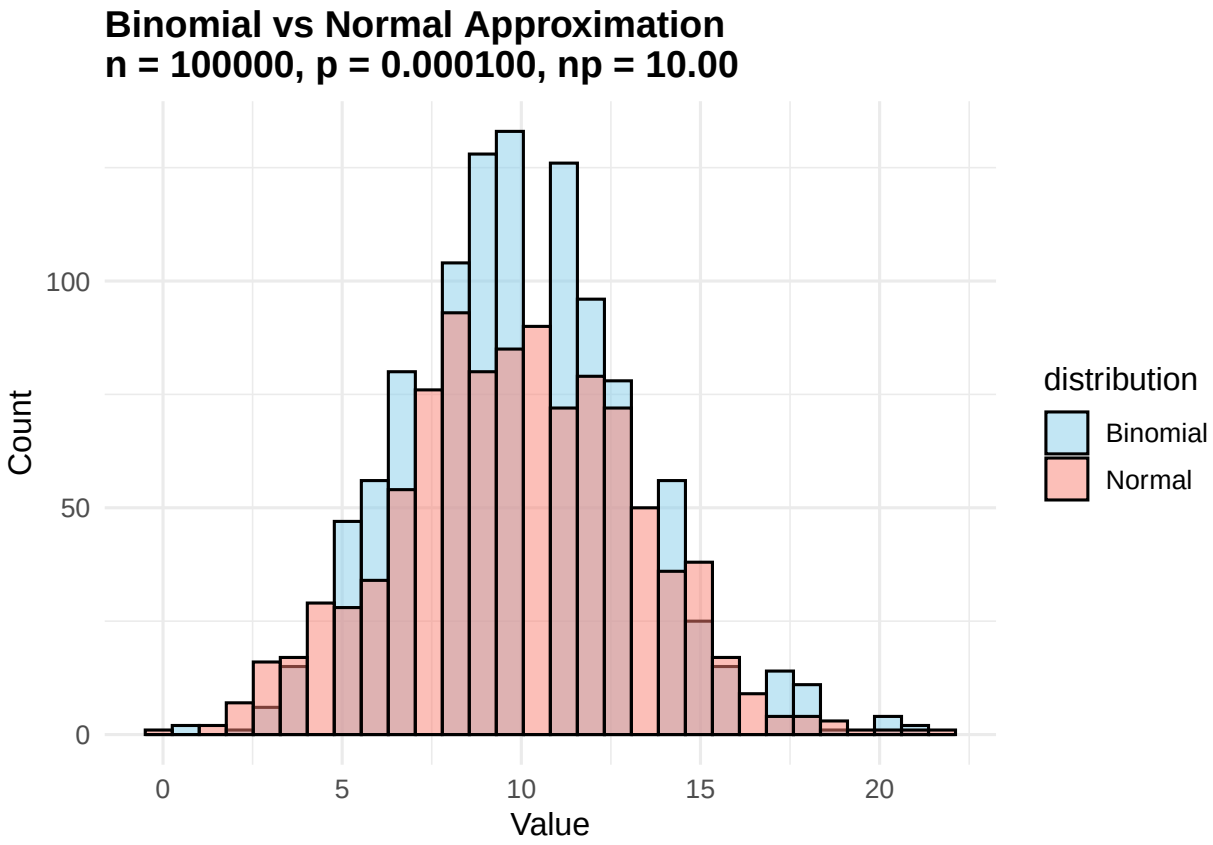

Binomial vs Normal Approximation
n = 100000, p = 0.031623, np = 3162.28
Count
90
60
30
0
3000
3100
3200
3300
Value
distribution
Binomial
Normal


Binomial vs Normal Approximation
n = 100000, p = 0.010000, np = 1000.00
Count
Value
100
75
50
25
0
900
950
1000
1050
1100
distribution
Binomial
Normal


Binomial vs Normal Approximation
n = 100000, p = 0.003162, np = 316.23
125
100
75
50
25
0
Count
250
300
350
Value
distribution
Binomial
Normal


Binomial vs Normal Approximation
n = 100000, p = 0.001000, np = 100.00
Count
90
60
30
0
80
100
120
140
Value
distribution
Binomial
Normal


Binomial vs Normal Approximation
n = 100000, p = 0.000316, np = 31.62
150
100
50
0
Count
20
30
40
50
Value
distribution
Binomial
Normal




From above, I found that $n > 1,000$ tends to give good results, and when $np = 10$ the plots don't quite look similar enough to be worth comparing.

So I want to use KL divergence to create a sort of heat-map of what values of $n$ and $p$ produce "good" approximations for the Normal to the Binomial.

```
library(entropy)

ns <- round(10^(seq(from = 0, to = 5, by = 0.5)))
ps <- 10^-(seq(from = 0.5, to = 3, by = 0.5))
ps <- sort(c(ps, 0.5, 1-ps))
nsamples <- 1000
nsim <- 100

results <- data.frame(
  n = integer(),
  p = numeric(),
  sim = integer(),
  KL = numeric()
)

set.seed(54321)

for (i in seq_along(ns)) {
  for (j in seq_along(ps)) {
    n <- ns[i]
    p <- ps[j]
```

```
    mu  <- n * p
    var <- n * p * (1 - p)

    for (sim_i in 1:nsim) {
      normal <- rnorm(n = nsamples, mean = mu, sd = sqrt(var))
      binom  <- rbinom(n = nsamples, size = n, prob = p)

      combined <- c(normal, binom)
      bins <- pretty(range(combined), n = 30)
      countsN <- hist(normal, breaks = bins, plot = FALSE)$counts + 1e-10
      countsB <- hist(binom, breaks = bins, plot = FALSE)$counts + 1e-10

      kl <- KL.plugin(countsB, countsN, unit = "log10")

      # Append result to data frame
      results <- rbind(results, data.frame(n = n, p = p, sim = sim_i, KL = kl))
    }
  }
}

# View first few rows
head(results)
```

```
##   n     p sim        KL
## 1 1 0.001   1 1.1938200
## 2 1 0.001   2 0.3815196
## 3 1 0.001   3 0.4040669
## 4 1 0.001   4 0.3757309
## 5 1 0.001   5 1.1191864
## 6 1 0.001   6 0.3818274
```

Now visualize results.

```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
KL.threshold = 0.25
summary_results <- results %>%
  group_by(n, p) %>%
  reframe(
    mean_KL = mean(KL),
```

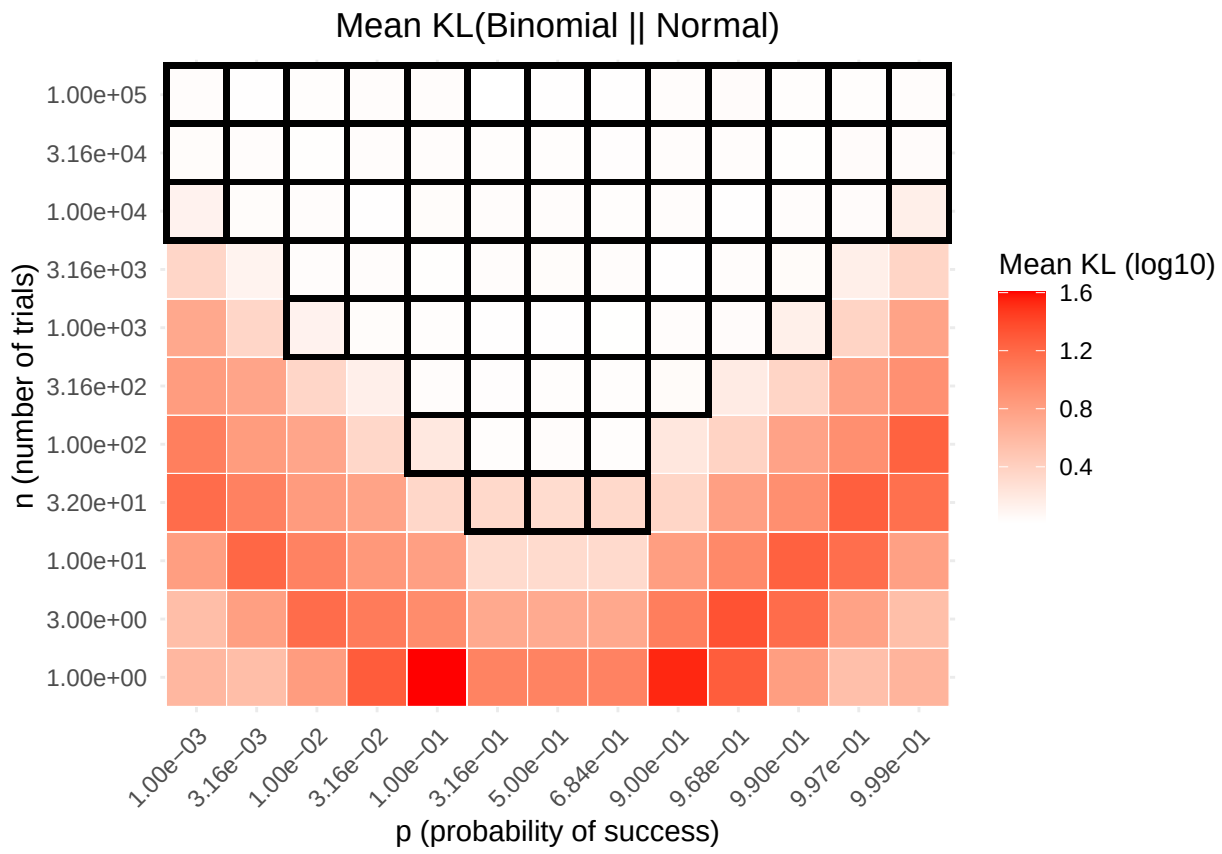
```
    var_KL = var(KL),
    threshold = mean(KL <= KL.threshold),
    np = n * p,
    np1 = n * (1 - p),
    valid_normal = (n * p >= 10) & (n * (1 - p) >= 10)
  )
df <- summary_results %>%
  mutate(
    n_f = factor(n, levels = sort(unique(n))),
    p_f = factor(p, levels = sort(unique(p)))
  )
```

```
sci3 <- function(x) {
  x_num <- as.numeric(as.character(x))
  format(x_num, scientific = TRUE, digits = 3)
}
```

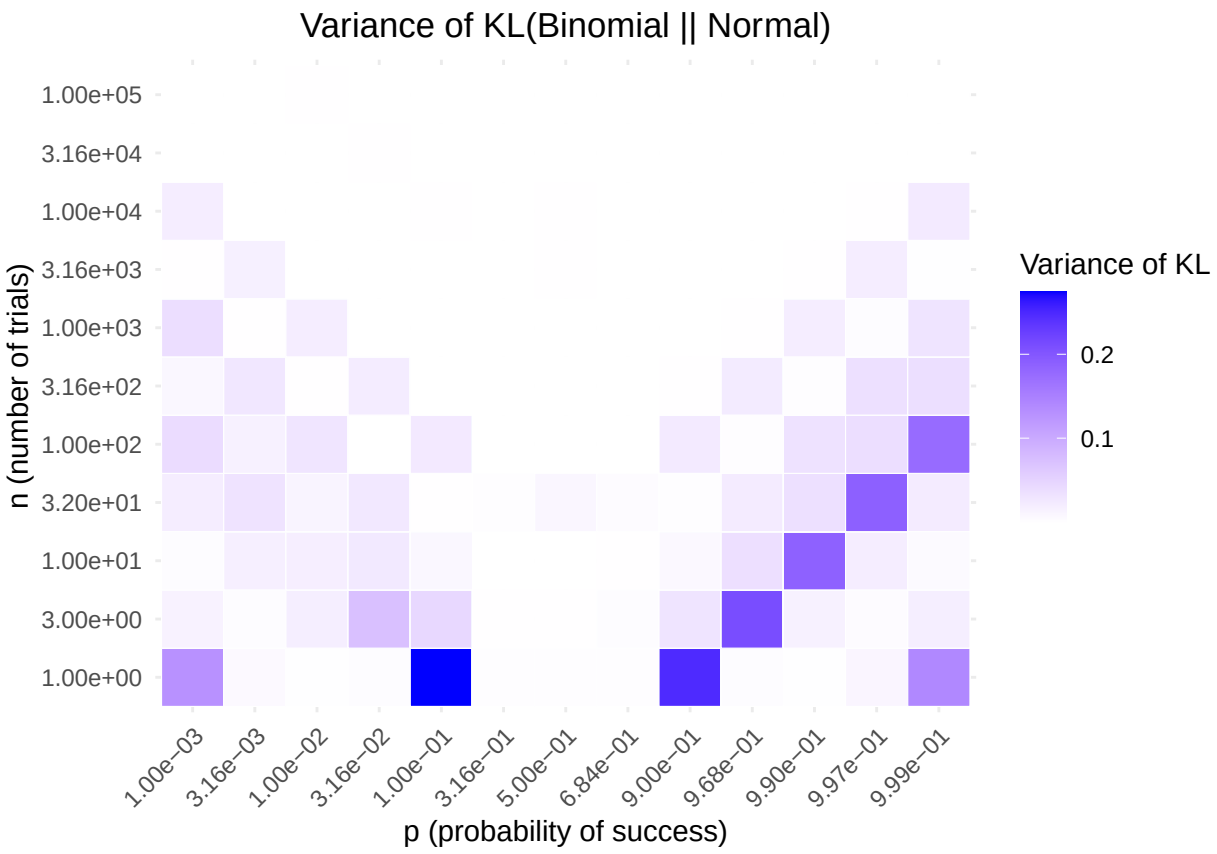
```
library(ggplot2)

ggplot(df, aes(x = p_f, y = n_f, fill = mean_KL)) +
  geom_tile(color = "white") +
  geom_tile(data = df[df$valid_normal, ],
            color = "black", size = 1, fill = NA) +  # Highlight valid cells
  scale_fill_gradient(low = "white", high = "red", name = "Mean KL (log10)") +
  labs(
    title = "Mean KL(Binomial || Normal)",
    x = "p (probability of success)",
    y = "n (number of trials)"
  ) +
  scale_x_discrete(labels = sci3) +
  scale_y_discrete(labels = sci3) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    plot.title = element_text(hjust = 0.5)
  )
```

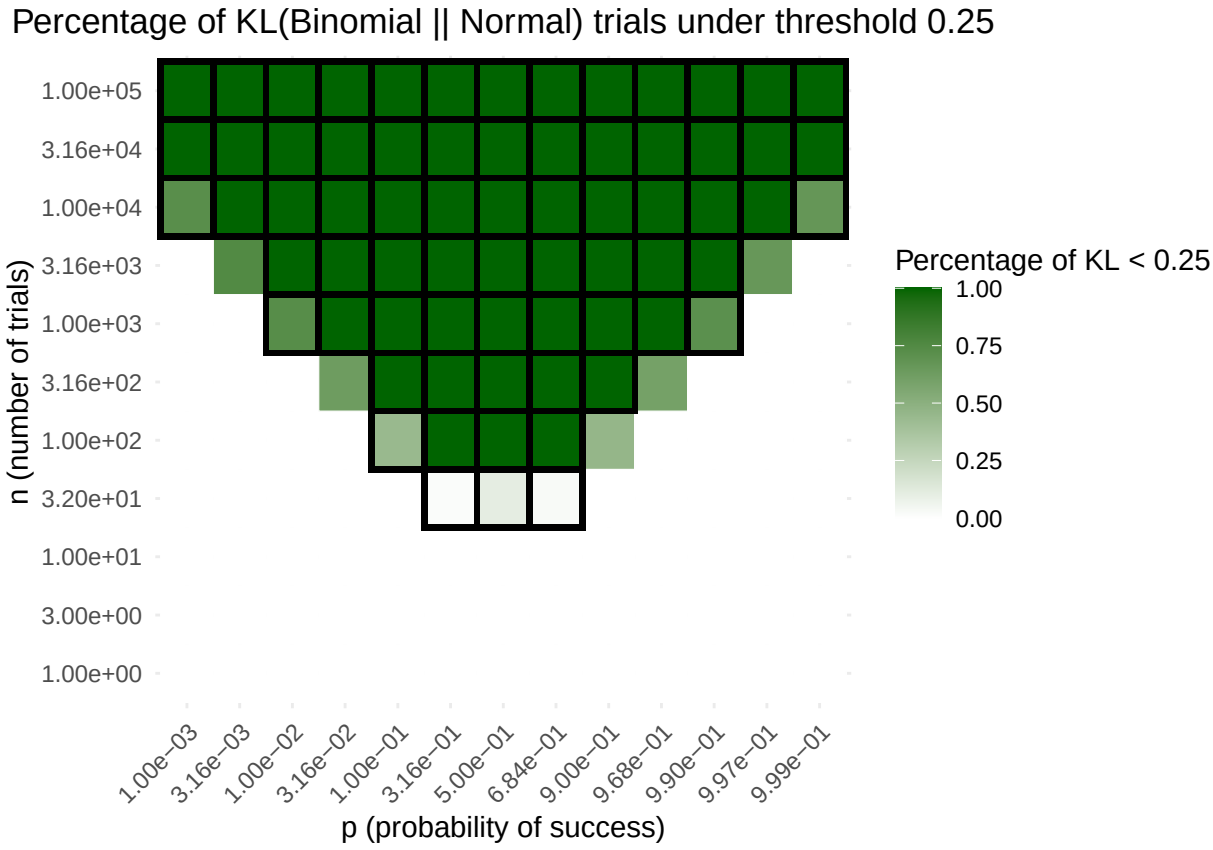
```
## Warning: Using ‘size‘ aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use ‘linewidth‘ instead.
## This warning is displayed once every 8 hours.
## Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
## generated.
```



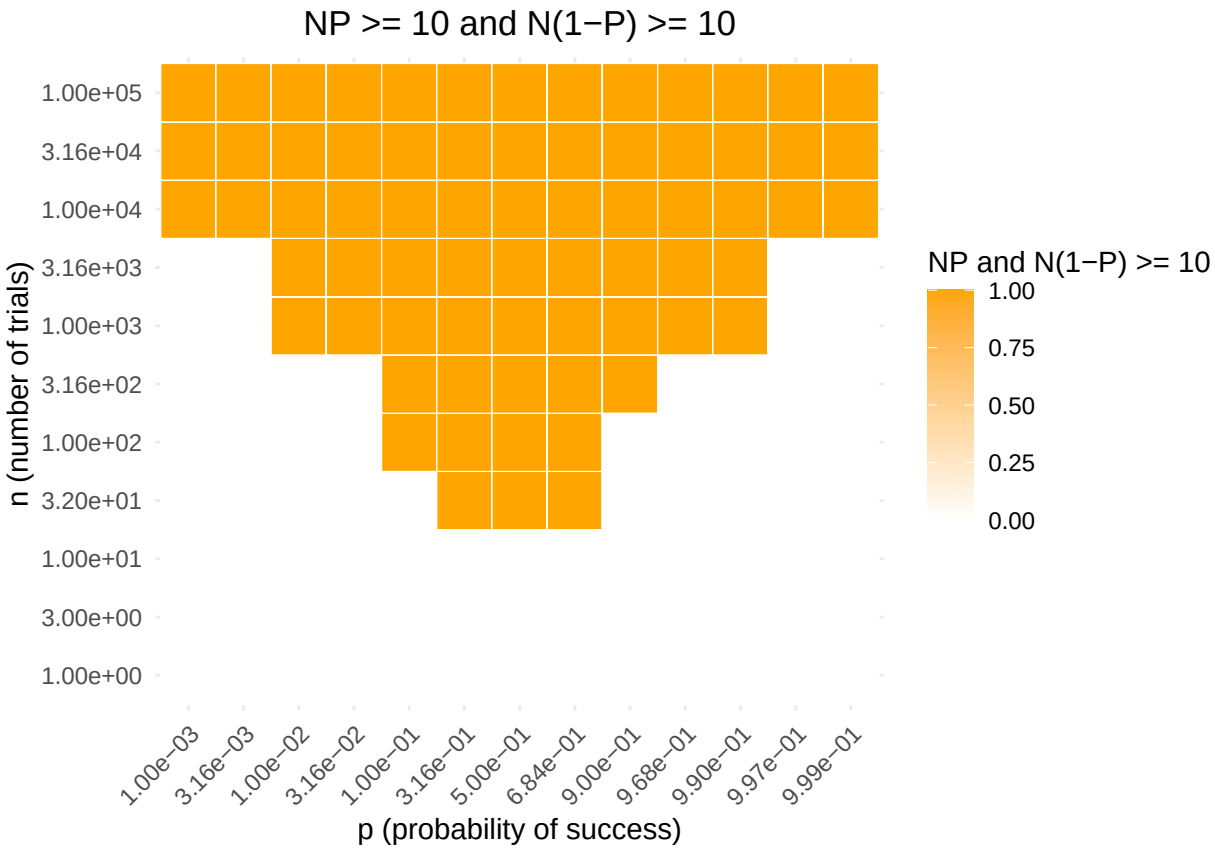

```r
ggplot(df, aes(x = p_f, y = n_f, fill = var_KL)) +
  geom_tile(color = "white") +
  #geom_tile(data = df[df$valid_normal, ],
  #          color = "black", size = 1, fill = NA) +  # Highlight valid cells
  scale_fill_gradient(low = "white", high = "blue", name = "Variance of KL") +
  labs(
    title = "Variance of KL(Binomial || Normal)",
    x = "p (probability of success)",
    y = "n (number of trials)"
  ) +
  scale_x_discrete(labels = sci3) +
  scale_y_discrete(labels = sci3) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    plot.title = element_text(hjust = 0.5)
  )
```




```r
ggplot(df, aes(x = p_f, y = n_f, fill = threshold)) +
  geom_tile(color = "white") +
  geom_tile(data = df[df$valid_normal, ],
            color = "black", size = 1, fill = NA) +  # Highlight valid cells
  scale_fill_gradient(low = "white", high = "darkgreen",
                      name = paste0("Percentage of KL < ", KL.threshold)) +
  labs(
    title = paste0("Percentage of KL(Binomial || Normal) trials under threshold ", KL.threshold),
    x = "p (probability of success)",
    y = "n (number of trials)"
  ) +
  scale_x_discrete(labels = sci3) +
  scale_y_discrete(labels = sci3) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    plot.title = element_text(hjust = 0.5)
  )
```




```r
ggplot(df, aes(x = p_f, y = n_f, fill = as.numeric(valid_normal))) +
  geom_tile(color = "white") +
  scale_fill_gradient(low = "white", high = "orange", name = "NP and N(1-P) >= 10") +
  labs(
    title = "NP >= 10 and N(1-P) >= 10",
    x = "p (probability of success)",
    y = "n (number of trials)"
  ) +
  scale_x_discrete(labels = sci3) +
  scale_y_discrete(labels = sci3) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    plot.title = element_text(hjust = 0.5)
  )
```


NP >= 10 and N(1−P) >= 10
n (number of trials)
1.00e+05
3.16e+04
1.00e+04
3.16e+03
1.00e+03
3.16e+02
1.00e+02
3.20e+01
1.00e+01
3.00e+00
1.00e+00
1.00e−03
3.16e−03
1.00e−02
3.16e−02
1.00e−01
3.16e−01
5.00e−01
6.84e−01
9.00e−01
9.68e−01
9.90e−01
9.97e−01
9.99e−01
p (probability of success)
NP and N(1−P) >= 10
1.00
0.75
0.50
0.25
0.00